

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing
Lecture 21 & 22 Scribe

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1 Chebyshev's Inequality

- **Core Concept:** It controls deviation from the mean by utilizing both mean and variance[cite: 139, 140, 141].
- **Formula:** $P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$ [cite: 143].
- **Proof Steps:** Apply Markov's inequality directly on $(X - \mu)^2$ [cite: 165].
- **Key Insight:** The bound decreases as the sample size n increases, making it significantly tighter than Markov's constant bound[cite: 190].

2 Chernoff Bound

- **Core Concept:** Uses the exponential transformation e^{tX} to amplify large tail values for a stricter bound[cite: 403, 404, 405, 406].
- **Upper Tail Formula:** $P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$ [cite: 504].
- **Lower Tail Formula:** $P(X \leq a) \leq \min_{t<0} \{ e^{-ta} E[e^{tX}] \}$ [cite: 713].
- **Steps to Solve:**
 1. Transform the base inequality using e^{tX} [cite: 496].
 2. Apply Markov's inequality to the transformed variables[cite: 497].
 3. Optimize the inequality by choosing the best t parameter to achieve the tightest bound[cite: 513].

3 Jensen's Inequality

- **Convexity Rule:** A function $g(x)$ is convex if $g(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha g(x_1) + (1 - \alpha)g(x_2)$ [cite: 757, 758].
- **Formula:** For a convex function $g(x)$ defined on an interval, $g(E[X]) \leq E[g(X)]$ [cite: 863, 864].
- **Interpretation:** Averaging the variables first results in a smaller value, while applying the convex function first yields a larger value[cite: 873, 874].

4 System Case Study: Server Crashing

- **Problem Statement:** Sudden, extreme traffic spikes cause server queues to overflow, resulting in system crashes and lost revenue[cite: 1092, 1093].
- **Model Setup:** Define Demand as X and Capacity as C . A system failure event occurs when $X \geq C$ [cite: 1101, 1102].
- **Steps to Scale Capacity:**
 1. **Markov Bound:** If telemetry only provides the average load, the system requires a massive safety fallback capacity (1,000,000 requests)[cite: 1252].
 2. **Chebyshev Bound:** Incorporating both average load and burstiness (variance) dramatically reduces the required capacity to 30,000 requests[cite: 1252].
 3. **Chernoff Bound:** Knowing the full probability distribution enables precise server node sizing at just 14,500 requests[cite: 1252].
- **The Golden Rule:** Adding more statistical information yields tighter bounds, which directly lowers hardware costs[cite: 1254, 1255].